BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Embedded Systems

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. When can additional CPU cores improve performance?

2. What makes an algorithm clear?

A) Steps in any order
B) Missing conditions
C) Instructions that need guessing
D) Precise steps in order

3. What does selection do?

4. Linear-search [7, 2, 9, 4] for 7, starting at index 0. List the values checked. How many comparisons are needed?

Today's objectives

Define an embedded system and recognise examples in everyday devices. Describe key characteristics of embedded systems using OCR J277 language.

Key vocabulary - Read, cover, recall

Embedded system
A computer built into a larger device to perform a specific function.

Dedicated function
A particular task for which a system has been designed.

Sensor
An input device that detects or measures a physical condition.

Actuator
An output device that produces a physical action, such as switching or moving a mechanism.

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Four key ideas

DEDICATED JOB	A computer built into a larger device for one purpose.
INPUT > PROCESS > OUTPUT	Sensor -> controller decision -> motor or signal.
SOFTWARE / MEMORY	Firmware in non-volatile memory; RAM for working data.
SUITABLE FEATURES	Often low power; some require a timely response.

I DO - WASHING-MACHINE CONTROLLER

INPUT	PROCESS	OUTPUT
Water-level sensor	Wash controller	Drum motor

A dedicated controller is built into a larger washing machine.

I Do - Embedded washing machine (4 marks)

A washing machine uses a processor to control its wash cycle. Explain why it is an embedded system, identify a sensor input, and name an output. [4 marks]

1. JOB: the controller performs the washing-cycle task.
2. EMBEDDED [2]: built into a larger device [1]; designed for a dedicated job [1].
3. INPUT [1]: water-level (or temperature) sensor sends a reading.
4. OUTPUT [1]: controller switches the drum motor on/off. Total = 4 marks.

We Do - Three guided questions (4 marks each)

BLACK PEN: identify the device / purpose / input / output.

1. Thermostat: why is the heating controller embedded? Identify its sensor input and output. [4]

2. Pedestrian crossing: explain the controller's dedicated job, push-button input and signal-light output. [4]

3. Fitness tracker: explain why low power and a limited interface suit a battery-powered device. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change temperature to 27. Does anything print?

```
temperature = 29
if temperature > 28 then
print("Ventilation on")
endif
```

BLACK PEN: complete the ONE short task on paper first.

TEST by tracing. IMPROVE by checking the full model on Page 4.

You Do - Independent exam questions

Q2. State TWO characteristics of an embedded system. [2 marks]

Q3. A car-park barrier uses a vehicle sensor and a motor to lift the barrier when authorised. Explain why its controller is an embedded system, and identify an input and an output. [4 marks]

Q4. A battery-powered smoke alarm uses a sensor to detect smoke and activates a loud sounder when smoke is detected. Explain THREE embedded-system characteristics that make it suitable for this job. [6 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. Every embedded system must have a full keyboard and monitor. ☐ T ☐ F
2. A thermostat may contain an embedded controller. ☐ T ☐ F
3. A sensor provides an input, not a physical output. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.

Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 More cores can help when work can be split between them, so tasks can make progress at the same time. ☐

Q2 D - Precise steps in order ☐

Q3 It chooses a branch using a condition. Selection can be nested. ☐

Q4 (a) [7]; 1 comparison(s), found at index 0. ☐

Code First - Model

```
temperature = 27
if temperature > 28 then
print("Ventilation on")
endif
```

EXPECTED: No output

☐

We Do - Three model answers

1. Thermostat: built into heating system [1]; dedicated heat control [1]; temperature sensor input [1]; switches heating output [1].

2. Crossing: built into crossing [1]; dedicated signals [1]; push-button input [1]; red/green lights output [1].

3. Tracker: low power [1] for longer battery life [1]; limited interface [1] is enough for short readings [1].

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

Explain one false statement using the key ideas.

You Do - Self-mark

Q2 Two features: built into a larger device [1] and a specific job [1]. ☐

Q3 Barrier system has a dedicated controller [2]; vehicle sensor is input [1]; motor is output [1]. ☐

Q4 Smoke alarm: dedicated smoke detection [2]; low power for battery [2]; real-time response for safety [2]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: